



WINTRIX ACADEMY PRESENTS

DCET Previous Year Question Paper & Solutions

2025

DIPLOMA COMMON ENTRANCE TEST

GUIDANCE • STRATEGY • SUCCESS

100 Questions • Detailed Solutions • Subject-Wise Organisation

FIRST EDITION — 2026



DCET 2025 Previous Year Question Paper & Solutions

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About Wintrix Academy

Wintrix Academy is a premier DCET mentorship platform founded with a singular mission: to empower Diploma students across Karnataka with guidance, strategy, and resources to secure top ranks.



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All-Karnataka DCET Rank 48 • 2nd Rank Hassan District • RVCE CSE (CGPA 9.19) • Placed at Red Hat

48

DCET RANK

9.8

DIPLOMA
CGPA

9.19

RVCE CGPA

50+

STUDENTS



Jnanesh Gowda

DIRECTOR & BRAND ARCHITECT

Turning confusion into clarity and effort into results through new-age learning strategies and results-engineered systems.

Preface

Dear DCET Aspirant,

This book contains the complete DCET 2025 Question Paper with detailed solutions for all 100 questions, organised subject-wise across five chapters.

The 2025 paper covers Engineering Mathematics, Statistics & Analytics, IT Skills, Electrical & Electronics Engineering, and Project Management Skills — each carrying 20 marks.

Attempt each question on your own first, then verify with the solution. Wishing you the very best!

Mohammed Adnan

Founder, Wintrix Academy

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01

CHAPTER ONE

Engineering Mathematics

20 Marks

Questions 1 – 20

Q.01

MATRICES

Choose the correct answer from the following:

- a) Every identity matrix is a scalar matrix
- b) Every scalar matrix is an identity matrix
- c) Every diagonal matrix is an identity matrix
- d) A square matrix whose each element is one is an identity matrix

✓ ANSWER: (A)

An identity matrix has 1's on the diagonal and 0's elsewhere. A scalar matrix has equal diagonal elements and 0's elsewhere. So every identity matrix is a scalar matrix (with scalar = 1).

Q.02

DETERMINANTS

The value of $\begin{vmatrix} 3 & 1 & -9 \\ -1 & 2 & 3 \\ 2 & 1 & -6 \end{vmatrix}$ is

- a) 45
- b) 90
- c) 0
- d) -45

✓ ANSWER: (C)

$$\begin{aligned} &= 3(-12-3) - 1(6-6) + (-9)(-1-4) \\ &= 3(-15) - 1(0) - 9(-5) \\ &= -45 - 0 + 45 = 0 \end{aligned}$$

Q.03

EIGENVALUES

The characteristic equation of $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ is

a) $\lambda^2 - 3\lambda - 2 = 0$

b) $\lambda^2 - 5\lambda + 10 = 0$

c) $\lambda^2 - 5\lambda - 2 = 0$

d) $\lambda^2 + 3\lambda + 10 = 0$

✓ ANSWER: (B)

$$|A - \lambda I| = 0$$

$$|(1-\lambda)(4-\lambda) - (-2)(3)| = 0$$

$$4 - \lambda - 4\lambda + \lambda^2 + 6 = 0$$

$$\lambda^2 - 5\lambda + 10 = 0$$

Q.04

MATRICES

If $|A| = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 \end{bmatrix}$, then $A \cdot \text{adj}(A)$ is equal to

a) $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 \end{bmatrix}$

b) $\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 \end{bmatrix}$

d) $\begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 \end{bmatrix}$

✓ ANSWER: (D)

For any square matrix, $A \cdot \text{adj}(A) = |A| \cdot I$. Given $|A| = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 \end{bmatrix}$, so $A \cdot \text{adj}(A) = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 \end{bmatrix}$.

Q.05

STRAIGHT LINES

The slope of the straight line joining two points $(3, -4)$ and $(-1, 5)$ is

a) $4/9$

b) $-4/9$

c) $-1/4$

d) $-9/4$

✓ ANSWER: (D)

$$m = (y_2 - y_1) / (x_2 - x_1) = (5 - (-4)) / (-1 - 3) = 9 / (-4) = -9/4$$

Q.06

STRAIGHT LINES

The angle of inclination of the straight line is $\pi/4$ and y-intercept is 3, then equation of the straight line is

a) $y = x + 3$

b) $y = x - 3$

c) $y = -x - 3$

d) $y = -x + 3$

✓ ANSWER: (A)

$$\theta = \pi/4 \rightarrow m = \tan(\pi/4) = 1, c = 3$$

$$y = mx + c = 1 \cdot x + 3 = x + 3$$

Q.07

STRAIGHT LINES

The equation of straight line passing through $(-5, 4)$ and parallel to X-axis is

a) $x - y + 9 = 0$

b) $x + 5 = 0$

c) $x + y + 1 = 0$

d) $y - 4 = 0$

✓ ANSWER: (D)

A line parallel to X-axis has slope $m = 0$. Passing through $(-5, 4)$: $y - 4 = 0$

Q.08

STRAIGHT LINES

What is the equation of the straight line passing through the points $(1, 5)$ and $(3, 2)$?

a) $7x - 4y + 13 = 0$

b) $3x + 2y - 17 = 0$

c) $3x + 2y - 13 = 0$

d) $3x + 2y + 13 = 0$

✓ ANSWER: (C)

$$m = (2-5)/(3-1) = -3/2$$

$$y - 5 = (-3/2)(x - 1)$$

$$2y - 10 = -3x + 3 \rightarrow 3x + 2y - 13 = 0$$

Q.09

The ramp in front of a gate is elevated by an angle of $\pi/3$ radians, then the angle in degrees is

- a) 15°
- b) 75°
- c) 60°
- d) 30°

✓ ANSWER: (C)

$$\text{Degrees} = \text{Radians} \times 180/\pi = (\pi/3) \times (180/\pi) = 60^\circ$$

Q.10

If $\sin(90^\circ - \theta) + \cos\theta = \sqrt{2} \cos(90^\circ - \theta)$, then the value of $\operatorname{cosec}\theta$ is

- a) $\sqrt{3/2}$ **b) $\sqrt{(3/2)}$**
- c) $\sqrt{(2/3)}$ d) $\sqrt{3}$

✓ ANSWER: (B)

$$\begin{aligned}\cos\theta + \cos\theta &= \sqrt{2} \sin\theta \\ 2\cos\theta &= \sqrt{2} \sin\theta \rightarrow \tan\theta = 2/\sqrt{2} = \sqrt{2} \\ \text{Opp} &= \sqrt{2}, \text{Adj} = 1, \text{Hyp} = \sqrt{3} \\ \operatorname{cosec}\theta &= \text{Hyp}/\text{Opp} = \sqrt{3}/\sqrt{2} = \sqrt{(3/2)}\end{aligned}$$

Q.11

TRIGONOMETRY

If A, B, C are the angles of a triangle, then the incorrect relation is

- a) $\sin((A+B)/2) = \cos(C/2)$
- b) $\cos((A+B)/2) = \sin(C/2)$
- c) $\tan((A+B)/2) = \sin(C/2)$**
- d) $\cot((A+B)/2) = \tan(C/2)$

✓ ANSWER: (C)

Since $A+B+C = 180^\circ$, $(A+B)/2 = 90^\circ - C/2$. So $\tan((A+B)/2)$ should equal $\cot(C/2)$, not $\sin(C/2)$. Option (c) is **incorrect**.

Q.12

TRIGONOMETRY

The value of $\sin 40^\circ + \sin 20^\circ$ is

- a) $\sqrt{3} \cos 10^\circ$
- b) $\sqrt{3} \cos 20^\circ$
- c) $\cos 10^\circ$**
- d) $\sqrt{2} \cos 10^\circ$

✓ ANSWER: (C)

$$\begin{aligned}\sin C + \sin D &= 2\sin((C+D)/2)\cos((C-D)/2) \\ &= 2\sin(30^\circ)\cos(10^\circ) \\ &= 2 \times (1/2) \times \cos 10^\circ = \cos 10^\circ\end{aligned}$$

Q.13

DIFFERENTIAL CALCULUS

The derivative of $y = (1+x)/(1-x)$ is

a) $2/(1-x)^2$

b) $-2/(1-x)^2$

c) $2/(1-x^2)$

d) $-2x/(1-x)^2$

✓ ANSWER: (A)

$$\begin{aligned} dy/dx &= [(1-x)(1) - (1+x)(-1)] / (1-x)^2 \\ &= (1-x+1+x) / (1-x)^2 = 2/(1-x)^2 \end{aligned}$$

Q.14

DIFFERENTIAL CALCULUS

If $y = \log(\sqrt{\tan x})$, then the value of dy/dx at $x = \pi/4$ is

a) ∞

b) 1

c) 0

d) $1/2$

✓ ANSWER: (B)

$$\begin{aligned} dy/dx &= (1/\sqrt{\tan x}) \times (1/2\sqrt{\tan x}) \times \sec^2 x \\ &= \sec^2 x / (2\tan x) \\ \text{At } x=\pi/4: (\sqrt{2})^2 / (2 \cdot 1) &= 2/2 = 1 \end{aligned}$$

If $y = (2 + 3 \log x)^{10}$, then dy/dx at $x = 1$ is

a) $15(2^9)$

b) $(15 \times 2)^{10}$

c) $15(2^{10})$

d) $10(5^9)$

✓ ANSWER: (C)

$$dy/dx = 10(2+3\log x)^9 \times (3/x)$$

$$\text{At } x=1 \text{ (log1=0)}: = 10(2)^9 \times 3 = 30(2^9)$$

$$= 15 \times 2 \times 2^9 = 15(2^{10})$$

If $y = \sqrt{(\sin x + y)}$, then $dy/dx =$

a) $\cos x / (2y - 1)$

b) $\cos x / (1 - 2y)$

c) $\sin x / (2y - 1)$

d) $\sin x / (1 - 2y)$

✓ ANSWER: (A)

$$y^2 = \sin x + y \text{ (squaring both sides)}$$

$$2y(dy/dx) = \cos x + dy/dx$$

$$dy/dx(2y - 1) = \cos x$$

$$dy/dx = \cos x / (2y - 1)$$

$\int \cot^2 x \, dx =$

a) $-\cot x - x + c$

b) $\cot x - x + c$

c) $\tan x - x + c$

d) $\tan^2 x + c$

✓ ANSWER: (A)

$$\cot^2 x = \operatorname{cosec}^2 x - 1$$

$$\int (\operatorname{cosec}^2 x - 1) dx = -\cot x - x + c$$

The value of $\int_0^{\pi/4} \sec^2 x / (1 + \tan x) \, dx$ is

a) $\log 2 + 1$

b) $\log 1$

c) $\log 2$

d) $\log 2 - 1$

✓ ANSWER: (C)

$$\text{Let } f(x) = 1 + \tan x, \quad f'(x) = \sec^2 x$$

$$\int f'(x)/f(x) \, dx = \log |f(x)|$$

$$= [\log(1 + \tan x)]_0^{\pi/4}$$

$$= \log(1 + 1) - \log(1 + 0) = \log 2 - 0 = \log 2$$

$\int \sqrt{1 + \cos 2x} \, dx$ is

- a) $2\cos x + c$ b) $2\sin x + c$
c) $\sqrt{2} \cos x + c$ d) $\sqrt{2} \sin x + c$

✓ ANSWER: (D)

$$\begin{aligned} 1 + \cos 2x &= 2\cos^2 x \\ \sqrt{2\cos^2 x} &= \sqrt{2} \cos x \\ \int \sqrt{2} \cos x \, dx &= \sqrt{2} \sin x + c \end{aligned}$$

The value of $\int_0^{\pi/4} \sin x \cos x \, dx$ is

- a) $-1/4$ b) $1/4$
c) $\pi/2$ d) $\pi/4$

✓ ANSWER: (B)

$$\begin{aligned} \int f(x)f'(x) \, dx &= [f(x)]^2/2 \\ f(x) &= \sin x, \quad f'(x) = \cos x \\ &= [\sin^2 x/2]_0^{\pi/4} = \sin^2(\pi/4)/2 - 0 \\ &= (1/\sqrt{2})^2/2 = (1/2)/2 = 1/4 \end{aligned}$$

02

CHAPTER TWO

Statistics & Analytics

20 Marks

Questions 21 – 40

Q.21

DATA

The data extracted from the Google forms has the file extension as

- a) .doc b) .docx
- c) .csv d) .xlsx

✓ ANSWER: (C)

Data from Google Forms is exported as **.csv** (Comma-Separated Values).

Q.22

DATA CLASSIFICATION

Match: a) Data–i) Survey, b) Questionnaire–ii) Number of students, c) Interview–iii) Google form, d) Quantitative data–iv) Facts and figure

- a)** a-iv, b-i, c-iii, d-ii
- b)** a-iv, b-ii, c-iii, d-i
- c)** a-iv, b-iii, c-i, d-ii
- d)** a-iv, b-i, c-ii, d-iii

✓ ANSWER: (C)

Data = facts & figures (iv), Questionnaire = Google form (iii), Interview = method of survey (i), Quantitative = numerically measurable like number of students (ii).

Data cleaning refers to

- a) Modifying the gathered data
- b) Adding data to the gathered data
- c) Removing the data from gathered data
- d) Processing the data according to the defined problem statement

✓ ANSWER: (A/B/C/D)

All options are objectives of **data cleaning**: modifying, adding, removing, and processing data.

The syntax to find frequency distribution in an interval in Excel is

- a) =COUNTIFS(criteria_range1, criteria1,)
- b) =COUNTIF(criteria_range1, criteria1,)
- c) =COUNT(criteria_range1, criteria1,)
- d) =COUNTIFS(criteria1, criteria2,)

✓ ANSWER: (A)

The syntax is **=COUNTIFS(criteria_range1, criteria1,)**

Which tab in Excel supports in plotting Pie chart?

- a) Home tab
- b) View tab
- c) Insert tab
- d) Data tab

✓ ANSWER: (C)

The **Insert tab** is used to plot pie charts in Excel.

Frequency polygon is a _____ graph

- a) Scatter graph
- b) Pie graph
- c) Bar graph
- d) Line graph

✓ ANSWER: (D)

A frequency polygon is a **line graph**.

Which of the following statements on Histogram is true?

- a) A Histogram can display categorical data effectively
- b) The bars in a Histogram should not touch each other
- c) A Histogram shows the frequency distribution of continuous data
- d) A Histogram is same as a bar chart

✓ ANSWER: (C)

A histogram shows **frequency distribution of continuous data**.

Choose the right stem-leaf plot for data: 52, 20, 44, 28, 20, 41, 30, 56, 32, 28

- a) Plot-1
- b) Plot-2
- c) Plot-3
- d) Plot-4

✓ ANSWER: (B)

Arranging data: 20,20,28,28,30,32,41,44,52,56. Stem|Leaf: 2|0088, 3|02, 4|14, 5|26. Key: 2|0=20.

PIE CHART

a) ₹30,000 b) ₹25,000

c) ₹15,000 d) ₹10,000

$$\begin{aligned} \text{Rent + Transport} &= 20\% + 10\% = 30\% \\ &= 30/100 \times 50000 = ₹15,000 \end{aligned}$$

STATISTICS

a) i-d, ii-c, iii-b, iv-a

b) i-d, ii-c, iii-a, iv-b

c) i-a, ii-b, iii-c, iv-d

d) i-d, ii-a, iii-b, iv-c

Range = $76 - 28 = 48$ (d)
Mean = $533 / 11 = 48.4545$ (c)
Mode = 58 (most frequent) (a)
Median = 44 (middle value) (b)

Q.31

EXCEL

The Excel syntax for finding quartile 4 (4th quartile) for a given data set is

- a) =QUARTILE [array, 4] **b) =QUARTILE (array, 4)**
- c) =QUARTILE (array_4) d) =QUARTILE (array-4)

✓ ANSWER: (B)

The syntax is =QUARTILE(array, 4)

Q.32

STATISTICS

Which equality is true for data: 25, 25, 25, 25, 25, 25, 25?

- a) Variance > Standard deviation **b) Variance = Standard deviation**
- c) Variance < Standard deviation d) Variance < 0, SD < 0

✓ ANSWER: (B)

All values are identical, so Variance = SD = 0. Hence Variance = Standard deviation .

Q.33

DISTRIBUTION

Skewness of the symmetric distribution shown is

- a) > 0 **b) = 0**
- c) < 0 d) = 1

✓ ANSWER: (B)

In a symmetric distribution, skewness $\beta_1 = 0$.

Q.34

KURTOSIS

If the Kurtosis = 3, then the data distribution is called

- a) Leptokurtic
- b) Mesokurtic
- c) Platykurtic
- d) Skewkurtic

✓ ANSWER: (B)

When $\beta_2 = 3$, the distribution is **mesokurtic** (normal distribution).

Q.35

PYTHON

Assertion (A): An "int" type data in Python can store floating point numbers.

Reasoning (R): Python "int" type only stores positive integers.

- a) A is true, R is false
- b) A is false, R is true
- c) A is false, R is false
- d) A is true, R is true

✓ ANSWER: (C)

A is false — `int` cannot store floats. R is also false — `int` stores positive, negative, and zero.

Q.36

PYTHON

Match data types: a) `a=5`, b) `a or b`, c) `"Hello KEA"`, d) `C=5.9` with i) string, ii) float, iii) Integer, iv) Boolean

- a) a-iii, b-iv, c-i, d-ii
- b) a-iv, b-i, c-ii, d-iii
- c) a-iv, b-i, c-iii, d-ii
- d) a-i, b-ii, c-iii, d-iv

✓ ANSWER: (A)

`a=5` → Integer(iii), `a or b` → Boolean(iv), `"Hello KEA"` → String(i), `C=5.9` → Float(ii)

Q.37

PYTHON

What is the syntax to assign the data type to a variable in Python?

- a) Name = "hello world" b) Height = 5.9
- c) Assertion = True **d) All of the above**

✓ ANSWER: (D)

Python uses **dynamic typing** — data types are assigned automatically based on the value.

Q.38

PYTHON

Which of the following is a valid variable name in Python?

- a) 1 Var **b) Var_name**
- c) Var - name d) Var name

✓ ANSWER: (B)

Var_name is valid — starts with letter, contains only letters and underscore, no spaces or special characters.

Q.39

PYTHON

Output of: n=5, A=0, while(n>0): A=A+n, n=n-1, print(A)

- a) 15** b) 5
- c) 55 d) 0

✓ ANSWER: (A)

```
n=5: A=0+5=5, n=4: A=5+4=9,  
n=3: A=9+3=12, n=2: A=12+2=14,  
n=1: A=14+1=15, n=0: loop stops.  
print(A) → 15
```

Which of the following is the correct syntax of an if statement in Python?

a) if x>0 then:

b) if (x>10)

c) if x>10:

d) if x>10;

✓ ANSWER: (C)

Python if syntax: `if condition:` — no parentheses needed, must end with colon.

03

CHAPTER THREE

IT Skills

20 Marks

Questions 41 – 60

Q.41

ALGORITHMS

Consider: Step 1: $A=5$, $B=3$. Step 2: if $A>B$, then $A=A-B$. Step 3: $B=A+B$. Step 4: Output B .
What is the output?

a) 2

b) 5

c) 8

d) 3

✓ ANSWER: (B)

$A=5$, $B=3$. $A>B$? Yes $\rightarrow A=5-3=2$

$B = A+B = 2+3 = 5$. Output $B = 5$

Q.42

ALGORITHMS

Which of the following best defines an algorithm?

a) A graphical representation

b) A code snippet

c) A step-by-step procedure to solve a problem

d) A mobile application

✓ ANSWER: (C)

An algorithm is a **finite sequence of well-defined steps** to solve a problem.

Q.43**ALGORITHMS**

Algorithm: if $a \leq b$ and $a \leq c \rightarrow$ print a; else if $b \leq a$ and $b \leq c \rightarrow$ print b; else $\rightarrow$ print c. If $a=10$, $b=25$, $c=33$, output?

a) 25

b) 10

c) 33

d) None of the above

✓ ANSWER: (B)

$10 \leq 25$ (True) and $10 \leq 33$ (True) $\rightarrow$ both conditions true $\rightarrow$ prints **10**.

Q.44**ALGORITHMS**

if $\text{num} > 0$: $\text{num} = \text{num} + 10$, print num; else if $\text{num} == 0$: $\text{num} = \text{num} * 10$, print num; else: print num. Output if $\text{num} = -5$?

a) 50

b) -50

c) -5

d) 5

✓ ANSWER: (C)

$-5 > 0$? No. $-5 == 0$? No. else: print num $\rightarrow$ prints **-5**.

Q.45**FLOWCHARTS**

Flowchart: Read a,b,c. If $a \leq b$ and $b \leq c \rightarrow$ print a; else if $b \leq a$ and $b \leq c \rightarrow$ print b; else $\rightarrow$ print c. What is output if $a=10$, $b=15$, $c=5$?

a) 10

b) 15

c) error

d) 5

✓ ANSWER: (D)

$a \leq b$ and $b \leq c \rightarrow 10 \leq 15$ (T) and $15 \leq 5$ (F) $\rightarrow$ False. $b \leq a$ and $b \leq c \rightarrow 15 \leq 10$ (F) $\rightarrow$ False. else $\rightarrow$ print c = **5**.

Which statement about web browsers is correct?

- a) Translate HTML, CSS, JS into machine code
- b) Can only display HTML and CSS, not execute JavaScript
- c) Interpret HTML, CSS & JavaScript to render a webpage**
- d) Use server-side languages like PHP to render directly

✓ ANSWER: (C)

Browsers **interpret HTML, CSS & JavaScript** to render webpages.

Consider: `<select name="cars"> <option>Volvo</option> <option>XUV</option> <option>Tata</option> </select>`. What does this do?

- a) Creates a text input field
- b) Creates a dropdown list for car brands**
- c) Allows multiple options from a list
- d) Creates a clickable button

✓ ANSWER: (B)

The `<select>` element creates a **dropdown list**.

Which CSS property is used to set the background color of a webpage?

- a) background-color**
- b) color
- c) bg-color
- d) background-image

✓ ANSWER: (A)

background-color sets the background color of an element.

Q.49

CSS

Which CSS selector is used to select an HTML element with a specific id?

- a) .id
- b) #id
- c) id
- d) [id]

✓ ANSWER: (B)

The hash symbol **#** is the ID selector in CSS.

Q.50

JAVASCRIPT

Which event is triggered when a user clicks on an HTML element in JavaScript?

- a) onload
- b) onclick
- c) onhover
- d) onchange

✓ ANSWER: (B)

The **onclick** event fires when a user clicks on an element.

Q.51

SEO

Which factor can help improve a website's SEO ranking?

- a) Using JavaScript for content rendering
- b) High Keyword density in the content
- c) Using relevant and high-quality backlinks
- d) Including as many images as possible

✓ ANSWER: (C)

Backlinks from reputable websites are "votes of confidence" for SEO.

Q.52**ORGANIZATION**

Correct descriptions of structured organisation: (a) Operations → produces goods/services
(b) Marketing → identifies customer needs (c) HR → manages budgeting and financial planning

a) (a), (b)

b) (a), (b), (c)

c) (a) only

d) (b) only

✓ **ANSWER: (A)**

HR does not handle budgeting/financial planning — that's Finance. Only **(a) Operations and (b) Marketing** are correct.

Q.53**ERP**

Which statement best describes the purpose of ERP tools in an organisation?

a) Manage customer relationships and marketing

b) Integrate and automate key business processes across departments

c) Primarily focused on managing email communication

d) Used for managing IT infrastructure and hardware

✓ **ANSWER: (B)**

ERP systems **unify operations across departments** for seamless coordination.

Q.54**CLOUD**

In cloud computing, data is stored in

a) Local folders

b) Physical notebooks

c) Remote servers accessed via the internet

d) USB drives

✓ **ANSWER: (C)**

Cloud computing stores data on **remote internet-connected servers**.

Which of the following is an example of IoT in smart homes?

- a) A GPS system in a car for navigation
- b) A thermostat that adjusts temperature based on occupancy and preferences**
- c) An e-commerce website for shopping
- d) A manual doorbell

✓ ANSWER: (B)

Smart **thermostats** use sensors and connectivity — a core IoT feature in smart homes.

Which of the following is an example of Software-as-a-Service (SaaS)?

- a) Microsoft Azure
- b) Google Drive
- c) Amazon Web Services (AWS) EC2
- d) Google Kubernetes Engine

✓ ANSWER: (A & B)

SaaS delivers applications over the internet. **Microsoft Azure** also provides SaaS, and **Google Drive** is a SaaS application.

Which of the following is a platform for developing IoT applications?

- a) Google Docs
- b) Arduino**
- c) Microsoft
- d) WinZip

✓ ANSWER: (B)

Arduino is an open-source electronics platform for building IoT devices.

Ravi uses incognito mode believing it protects from all tracking and threats. Which statement is correct?

- a) Provides full malware protection
- b) Hides activity from device and all trackers
- c) Only prevents local history tracking, not online threats**
- d) Encrypts his traffic and login credentials

✓ ANSWER: (C)

Incognito mode only stops saving local history, cookies, and site data. It does **not protect** from malware or phishing.

Rita uses a public computer at a cyber cafe for email and closes the browser without logging out. What cyber safety mistake did she make?

- a) Read personal emails in public
- b) Used a public computer
- c) She didn't logout of her account**
- d) She didn't clear browsing history

✓ ANSWER: (C)

Not logging out on public computers poses a risk of unauthorized access.

To safeguard against cyber attacks that steal sensitive information, which browser setting should be enabled?

- a) Enable automatic updates
- b) Turn on phishing and malware protection**
- c) Disable phishing and malware protection
- d) Enable cookies

✓ ANSWER: (B)

Phishing and malware protection warns users about harmful websites and downloads.

04

CHAPTER FOUR

Fundamentals of Electrical & Electronics Engineering

20 Marks

Questions 61 – 80

Q.61

SYMBOLS

Match: i) Resistance, ii) Transformer, iii) Capacitor, iv) Battery with their circuit symbols a, b, c, d

a) i-a, ii-c, iii-d, iv-b

b) i-d, ii-b, iii-c, iv-a

c) i-b, ii-c, iii-d, iv-a

d) i-d, ii-c, iii-a, iv-b

✓ ANSWER: (D)

Resistance → zigzag, Transformer → coils, Capacitor → parallel lines, Battery → long/short lines.

Q.62

EARTHING

Earthing plate is made up of

a) Bakelite

b) Aluminum

c) Copper

d) Plastic

✓ ANSWER: (C)

Copper is a good conductor and corrosion-resistant, ideal for earthing.

Q.63

UNITS

Match: i) Current, ii) Power, iii) Energy, iv) Frequency with a) Watts, b) Hertz, c) kWh, d) Ampere

a) i-a, ii-b, iii-d, iv-c

b) i-a, ii-c, iii-b, iv-d

c) i-c, ii-b, iii-d, iv-a

d) i-d, ii-a, iii-c, iv-b

✓ ANSWER: (D)

Current → Ampere, Power → Watts, Energy → kWh, Frequency → Hertz.

CIRCUITS

a) $18\ \Omega$

b) 06 Ω

c) 04Ω

d) 02Ω

$$1/R = 1/12 + 1/6 = (1+2)/12 = 3/12 = 1/4$$
$$R = 4 \, \Omega$$

OHM'S LAW

a) 72A

b) 0.5A

c) $2A$

d) 1.2A

$$I = V/R = 12/6 = 2A$$

AC THEORY

a) 100 Hz

b) 200 Hz

c) 50 Hz

d) 150 Hz

$$f = 1/T = 1/0.02 = 50 \text{ Hz}$$

Q.67**AC THEORY**

In a pure resistive circuit, which statement is true?

- a) Both current and voltage are in phase
- b) Current leads the voltage
- c) Voltage lags the current
- d) Both are out of phase

✓ ANSWER: (A)

In a pure resistive AC circuit, current and voltage are **in phase**.

Q.68**PROTECTION**

HRC fuses can be used up to a maximum current of

- a) 250 mA
- b) 500 mA
- c) 1 A
- d) 1250 A

✓ ANSWER: (D)

HRC (High Rupturing Capacity) fuses handle very high fault currents, up to **1250 A or more**.

Q.69**WIRING**

To control a single lamp from two different places, the switch used is

- a) Double pole
- b) Two way
- c) Single pole
- d) One way

✓ ANSWER: (B)

Two-way switches allow control from two different locations.

Q.70**FUSE**

A fuse is always connected _____ to the circuit.

- a) Parallel
- b) Across the load
- c) Series**
- d) Across the supply

✓ ANSWER: (C)

A fuse is connected in **series** to break the circuit on overload.

Q.71**TRANSFORMER**

In a Transformer, _____ remains constant

- a) Voltage
- b) Frequency**
- c) Current
- d) Both current and voltage

✓ ANSWER: (B)

In both step-up and step-down transformers, **frequency** remains the same (e.g., 50 Hz).

Q.72**BATTERIES**

The combination of two or more cells is

- a) Battery**
- b) load cell
- c) Primary cell
- d) Generator

✓ ANSWER: (A)

A **battery** is formed by connecting two or more cells.

Q.73

MATCHING

Match: i) Ampere hour, ii) KVA, iii) Star delta, iv) Fan with a) 1 ϕ Induction motor, b) lead acid battery, c) Transformer, d) 3 ϕ Induction motor

- a) i-a, ii-b, iii-c, iv-d b) i-c, ii-a, iii-d, iv-b
- c) i-b, ii-c, iii-d, iv-a** d) i-b, ii-a, iii-c, iv-d

✓ ANSWER: (C)

Ampere-hour → Battery, KVA → Transformer, Star-delta → 3 ϕ motor, Fan → 1 ϕ motor.

Q.74

TRANSFORMER

In a transformer, the winding to which the load is connected is called _____ winding

- a) Primary b) Axillary
- c) Secondary** d) Concentric

✓ ANSWER: (C)

The **secondary winding** is connected to the load and delivers output voltage.

Q.75

TRANSFORMER

Which statement is FALSE with respect to transformer?

- a) Converts high voltage to low voltage
- b) Converts high frequency to low frequency**
- c) Converts low voltage to high voltage
- d) Transfers energy at constant frequency

✓ ANSWER: (B)

A transformer does **not change frequency**. It can step up or step down voltage, but frequency remains unchanged.

Q.76**LOGIC GATES**

Match logic gates with their symbols: i) AND, ii) OR, iii) NAND, iv) NOR

- a) i-a, ii-b, iii-c, iv-d b) i-b, ii-c, iii-d, iv-a
- c) i-d, ii-a, iii-c, iv-b **d) i-c, ii-d, iii-a, iv-b**

✓ ANSWER: (D)

AND, OR, NAND, NOR matched with their respective standard symbols.

Q.77**SENSORS**

Which is a common type of sensor used for measuring temperature?

- a) Proximity sensor b) Pressure sensor
- c) Flow sensor **d) Thermocouple**

✓ ANSWER: (D)

A **thermocouple** converts thermal energy into electrical voltage for temperature measurement.

Q.78**RESISTORS**

What does the fourth band in a 4-band resistor indicate?

- a) Resistance b) Multiplier
- c) Power rating **d) Tolerance**

✓ ANSWER: (D)

1st & 2nd bands = significant digits, 3rd = multiplier, 4th = **tolerance**.

Which of the following is an example of Insulator?

- a) Aluminium
- b) Rubber
- c) Iron
- d) Silicon

✓ ANSWER: (B)

Rubber resists the flow of electric current. Aluminium and Iron are conductors; Silicon is a semiconductor.

Diode is made up of _____ material.

- a) Conducting
- b) Semiconducting
- c) Insulating
- d) Electronic

✓ ANSWER: (B)

Diodes are made from **semiconductor** materials like Silicon or Germanium.

05

CHAPTER FIVE

Project Management Skills

20 Marks

Questions 81 – 100

Q.81**ANALOGY**

Complete the analogy: Project : Deliverables :: Operation : _____

- a) Deadlines
- b) Efficiency
- c) Repetition**
- d) Conclusion

✓ ANSWER: (C)

Projects have specific deliverables; operations are permanent processes with **repetitive** outcomes.

Q.82**OPERATIONS**

Multiple objectives are achieved again and again in _____

- a) Operation**
- b) Project
- c) Business
- d) Sales

✓ ANSWER: (A)

Operations are repetitive processes where objectives are achieved again and again.

Q.83**CONSULTANCY**

Which are examples of consultancy firms? A) Accenture, B) TCS, C) Tata Motors, D) Birla Technical Services

- a) A, B and C only
- b) A, B and D only**
- c) A and D only
- d) A, B, C and D

✓ ANSWER: (B)

Accenture, TCS, and Birla Technical Services are consultancy firms. **Tata Motors** is automotive manufacturing.

Which of the following is NOT an obstacle in project management?

- a) Undefined project scope
- b) Poor communication among team members
- c) Clear project objectives and goals**
- d) Limited resources

✓ ANSWER: (C)

Clear project objectives is an advantage, not an obstacle.

Project Procedure Manual : Standard Operating Procedures :: Project Diary : _____

- a) Records daily events and issues**
- b) Provides step-by-step instructions
- c) Outlines rules and processes
- d) Used only during project closure

✓ ANSWER: (A)

A project diary **records daily events, activities, and issues** throughout the project life cycle.

Match: i) Initial project team → d) Helps start the project; ii) Core project team → b) Works daily; iii) Stakeholders → c) Interested in outcomes; iv) Advisors → a) Guides with advice

- a) i-d, ii-b, iii-c, iv-a**
- b) i-a, ii-b, iii-c, iv-d
- c) i-b, ii-a, iii-c, iv-d
- d) i-d, ii-c, iii-b, iv-a

✓ ANSWER: (A)

Initial team starts the project (d), Core team works daily (b), Stakeholders are interested parties (c), Advisors guide with advice (a).

What does Work Breakdown Structure (WBS) do to help a Project Manager?

- a) Track the project's financial performance
- b) Breakdown the project into smaller, manageable tasks**
- c) Determine project's risk level
- d) Allocate resources to team members

✓ ANSWER: (B)

WBS **breaks down complex work into smaller, manageable tasks**.

Which is NOT a subpart of Project Execution Plan (PEP)?

- a) Organizational plan
- b) Work packing plan
- c) Staffing plan**
- d) Contracting plan

✓ ANSWER: (C)

Staffing plan is not included in PEP.

High demand, high selling price, low variable cost are examples of

- a) Normal Scenario
- b) Worst Scenario
- c) Best Scenario**
- d) Abnormal Scenario

✓ ANSWER: (C)

High demand, high selling price, low cost represent the **best scenario**.

In _____ phase, project definition and problem identification are defined

- a) Planning
- b) Execution
- c) Initiation**
- d) Closure

✓ ANSWER: (C)

Initiation is the first phase where project is defined and problems are identified.

Risk arising due to fluctuation of interest rate OR mistakes in accounting is known as _____ risks.

- a) Policy
- b) Technical
- c) Marketing
- d) Financial**

✓ ANSWER: (D)

Financial risks arise from interest rate fluctuations or accounting errors.

Assertion (A): Time overrun can lead to cost overrun in a project.

Reason (R): More time often means more money spent on labour and resources.

- a) Both (A) and (R) are true and (R) is the correct explanation of (A)**
- b) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- c) (A) is true, but (R) is false
- d) (A) is false, but (R) is true

✓ ANSWER: (A)

Both statements are true, and the reason correctly explains why time overrun leads to cost overrun.

Q.93

CPM

The full form of CPM in project management is

- a) Critical Process Management
- b) Critical Path Method**
- c) Continuous Project Monitoring
- d) Cost Planning Model

✓ ANSWER: (B)

CPM stands for **Critical Path Method**.

Q.94

TOOLS

Which is NOT a tool of project planning, scheduling and monitoring?

- a) Gantt Chart
- b) Flow chart**
- c) PERT Chart
- d) Milestone Chart

✓ ANSWER: (B)

Flow chart is not a project planning/scheduling/monitoring tool.

Q.95

EVALUATION

Project Evaluation can be done _____

- a) At the end of project only
- b) Throughout the project lifecycle**
- c) Only during planning phase
- d) At the start only

✓ ANSWER: (B)

Project evaluation is done **throughout the project lifecycle**.

Identify the incorrect statement regarding project planning functions:

- a) Defines project goals and objectives
- b) Allocates resources without considering constraints**
- c) Establishes a project timeline and milestones
- d) Identifies potential risks and mitigates them

✓ ANSWER: (B)

Project planning allocates resources **by considering constraints**, not without.

Which is NOT a key component of "SMART" goal?

- a) Time-Bound
- b) Random**
- c) Achievable
- d) Specific

✓ ANSWER: (B)

SMART = Specific, Measurable, Achievable, Realistic, Timely. **Random** is not a component.

The process of correcting and rectifying the errors in the project is

- a) Project Audit
- b) Project Monitoring
- c) Project Evaluation
- d) Project Review**

✓ ANSWER: (D)

Project Review is the process of correcting and rectifying errors.

Optimistic Time = 4 days, Pessimistic Time = 8 days, Most likely time = 6 days. The expected time to complete is

- a) 7 days
- b) 6 days
- c) 5 days
- d) 4 days

✓ ANSWER: (B)

$$T_e = (t_o + 4t_m + t_p)/6$$
$$= (4 + 4(6) + 8)/6 = 36/6 = 6 \text{ days}$$

Watching 3D movie is an example of

- a) Augmented Reality
- b) Virtual Reality
- c) Augmented and Virtual Reality
- d) Cloud Technology

✓ ANSWER: (C)

Watching a 3D movie is an example of both **augmented and virtual reality**.



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